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ALLOMETRY AND JUMPING IN FROGS: HELPING THE TWAIN TO MEET

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Superficially, anurans appear stereotyped in body shape, with short trunks and relatively long hindlimbs. This uniformity has been characterized by Inger (1967), who, borrowing from Gertrude Stein, wrote *a frog is a frog is a frog*. Because frogs preserve a generally similar body shape over a wide size range, they are an interesting group in which to study the relationship of locomotor function to body geometry and size.

On land, jumping is the dominant locomotor mode of anurans. Previous studies of frog jumping (e.g., Wermel, 1934; Rand and Rand, 1966; Zug, 1972) have assumed that the principal function of this locomotor mode is to maximize distance between the animal and a potential predator, and have focused on the relationship between morphology and jump distance. There are, however, other parameters of jumping, such as quickness of movement (acceleration) that may be important and related to frog morphology. A correlated examination of allometry and locomotor behavior provides an approach for testing two alternative hypotheses of jump function: jump distance vs. quickness of movement.

MODELS OF FORM AND FUNCTION

Hill (1950) suggested, on theoretical grounds, that animals which are geometrically similar may be able to jump the same absolute distance regardless of size. Rapid acceleration over a short time in small animals would result in the same distance that larger animals would attain through slower acceleration acting over a longer time. If Hill's model is correct, and if frogs are geometrically similar regardless of size, then small frogs should be able to jump as far as large frogs. Thus one possible function—providing a specific jump distance—would be preserved with changes in size. However, a second function—providing a specific quickness of movement (acceleration)—would be diminished with increasing size, because larger animals have lesser accelerations. For this model (a model of morphological isometry), the relationship between two possible jump functions (distance and acceleration) and size (body weight) can be represented graphically (Figs. 1*a* and *b*).

Maintenance of geometric similarity imposes limitations on acceleration and jump distance with increasing size. But, if geometric similarity is not maintained, acceleration may be maintained constant and jump distance may be increased with increasing size. It would be possible, for example, for relative jump distance (i.e., absolute distance/body weight) to be maintained constant with increasing body size. These alternative relationships between jump functions and size are represented in Figs. 1*c* and 1*d*, and are predictions of a morphological allometry model. The special case of same relative jump distance with increasing size is diagrammed (Fig. 1*c*, $a = 1$), but the general prediction of the morphological allometry model is only that there will be some change in distance with increasing size ($a \neq 0$).

These viewpoints provide a framework for testing alternative hypotheses of jump

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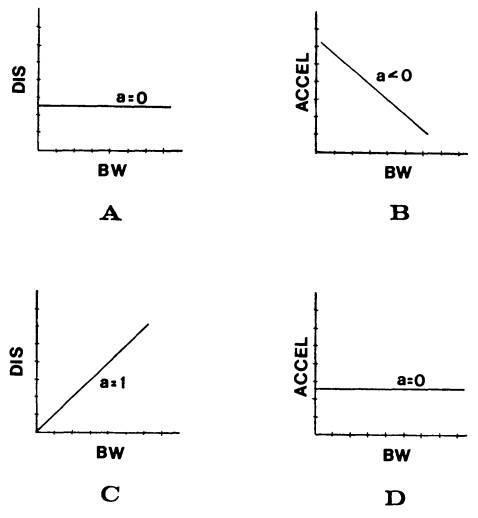


FIG. 1. Predictions of the morphological isometry model ($1A + 1B$). Predictions of the morphological allometry model ($1C + 1D$). a = slope, BW = body weight, dis = max. jump, accel. = acceleration.

function by measuring size, shape, and locomotor parameters in frogs and comparing the results with the predictions of the morphological isometry and allometry models. If the principal function of jumping is to provide a specific absolute jump distance, and if Hill's model is correct, then one would predict that with increasing size, body proportions and jump distance would remain constant and acceleration would decrease. If the principal function of jumping is quickness of movement or maintenance of the same relative jumping distance, then one would predict that with increasing size body shape would change, larger frogs would jump longer distances than smaller frogs, and acceleration would remain constant.

METHODS AND MATERIALS

The morphological similarity model assumes that frogs maintain geometric similarity with increase in body size. The morphological allometry model assumes that frogs change body shape with increase in size. These assumptions were tested by measuring body weight, body length, length of the hindlimbs, and hindlimb extensor muscle weight in size series of frog species from museum collections (Appendix I). Weights of preserved animals are generally considered to be unreliable, but in anurans they are no more variable than weights of live animals. Individual live frogs can vary up to 20% in body weight over a 12-h period (Emerson, unpublished observations). Even with differences in preservation, if anuran museum specimens are slit to remove excess preservative and blotted dry before weighing, the difference between preserved weight and live weight of a specimen is less than the possible variability in live weight of an individual frog over a 24-h period. Live weight is important in this study only for calculating locomotor parameters from trajectory formulae, and in that part of the study (see below), weights from live animals were used and the animals were weighed directly after each jump sequence.

All measurements were taken on the right side of the animal with a vernier caliper (0.01-mm precision); measurement definitions of body weight, body length and hindlimb length are those of Zug (1972). The principal hindlimb extensor muscles are those defined by Calow and Alexander (1973): the *plantaris longus*, *cruralis*, *gluteus magnus*, *semimembranosus*, *gracilis major*, and *semitendinosus*. Muscle weight was determined by excising the appropriate muscles, drying them for 48 h at 105 C, and then immediately weighing them on a Mettler balance. Regression analysis involved use of the method of the least squares. Confidence intervals were calculated for all pairs of variables. A Student's t-test was used to test for significance of differences between two regression coefficients. Regression axes were calculated by the standard power function of allometry:

$$\log y = a \log x + \log b \quad (y = bx^a).$$

Ontogenetic series of *Pseudacris triseriata*, *Bufo americanus*, and *Rana pipiens*

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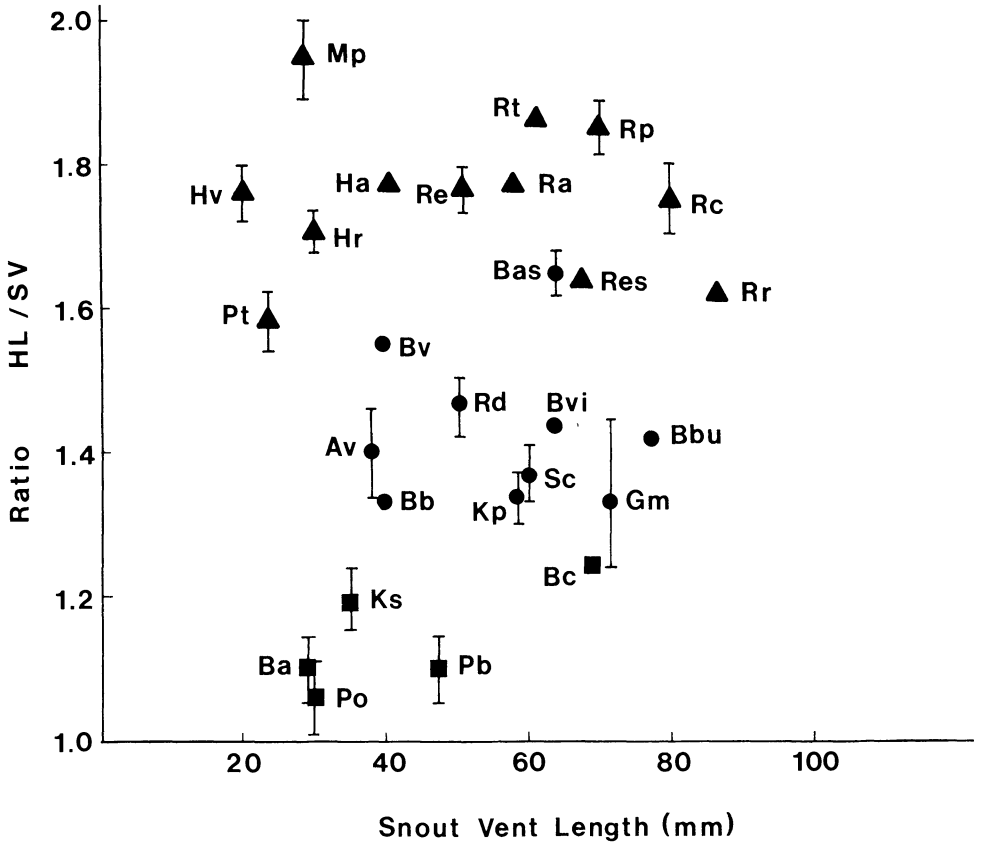


FIG. 2. Graph of mean hindlimb:snout-vent ratios vs. mean snout-vent length for adults of 27 species of frogs. Walkers (squares): Bc, *Bufo calamita*; Ba, *Breviceps adspersus*; Pb, *Phrynomerus bifasciata*; Po, *Pseudophryne occidentalis*; Ks, *Kassina senegalensis*. Hoppers (circles): Kp, *Kaloula pulchra*; Gm, *Glyphoglossus molossus*; Bbu, *Bufo bufo*; Bvi, *Bufo viridis*; Sc, *Scaphiopus couchi*; Rd, *Rhinophrynus dorsalis*; Av, *Atelopus varius*; Bas, *Bufo asper*; Bb, *Bombina bombina*; Bv, *Bombina variegata*. Jumpers (triangles): Pt, *Pseudacris triseriata*; Res, *Rana esculenta*; Rr, *Rana ridibunda*; Ra, *Rana arvalis*; Re, *Rana erythracea*; Rt, *Rana temporaria*; Rp, *Rana pipiens*; Ha, *Hyla arborea*; Hr, *Hyla regilla*; Hv, *Hyperolius viridiflavius*; Mp, *Microhyla pulchra*; Rc, *Rana catesbeiana*. Single points represent data from Dobrowolska (1973). Data points with 95% confidence intervals are from this study.

were used in jumping experiments to quantify differences in locomotor behavior with variation in size and shape.

Adults and juveniles of *B. americanus* and *P. triseriata* were collected from wild populations in southern Cook County, Illinois. Eggs from wild populations of both species were raised in the laboratory to provide a supply of the "just-metamorphosed" size class. In addition to *P. triseriata* from Illinois, adults of *P. triseriata* from the Rocky Mountain Field Laboratory, Colorado (a short legged form of the species) were included in the experiments. All *R. pipiens* were obtained from the Amphibian Facility, University of Michigan.

Jumping tests were performed at 22 C using the experimental procedure described by Zug (1972). Each animal was jumped 10 times in an experiment. Experiment days were divided into four periods: 6–12 A.M., 12–6 P.M., 6–12 P.M., and 12–6 A.M.,

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TABLE 1. *The mean hindlimb/snout vent ratio for adult frogs of varying locomotor modes and ecology.*

Family	Species	No.	HL/SV ratio	95% Con. interval	Locomotion/ecology
Leptodactylidae	*Pseudophryne occidentalis*	5	1.07	±.06	walking, burrowing/terrestrial[1]
Microhylidae	*Phrynomerus bifasciata*	12	1.11	±.04	running/terrestrial[2]
Microhylidae	*Breviceps adspersus*	26	1.14	±.04	walking, burrowing/terrestrial[2]
Ranidae	*Kassina senegalensis*	10	1.19	±.04	running/terrestrial[2]
Microhylidae	*Glyphoglossus molossus*	10	1.30	±.12	walking, burrowing/terrestrial[3]
Microhylidae	*Kaloula pulchra*	21	1.32	±.04	walking, burrowing/terrestrial[3]
Pelobatidae	*Scaphiopus couchi*	12	1.37	±.04	hopping, burrowing/terrestrial[3]
Bufonidae	*Atelopus varius*	13	1.41	±.08	walking, jumping/terrestrial[4]
Rhinophrynidae	*Rhinophrynus dorsalis*	10	1.47	±.04	walking, burrowing/terrestrial[3]
Hylidae	*Pseudacris triseriata*	24	1.64	±.04	jumping/terrestrial[3]
Bufonidae	*Bufo asper*	38	1.65	±.03	hopping/terrestrial[5]
Hylidae	*Hyla regilla*	27	1.71	±.03	jumping/semi-arboreal[6]
Ranidae	*Rana catesbeiana*	18	1.75	±.05	jumping/aquatic[6]
Ranidae	*Hyperolius viridiflavius*	18	1.76	±.04	jumping/arboreal[3]
Ranidae	*Rana erythracea*	28	1.78	±.03	jumping/terrestrial[5]
Ranidae	*Rana pipiens*	25	1.85	±.04	jumping/terrestrial[6]
Microhylidae	*Microphyla pulchra*	12	1.94	±.06	jumping/terrestrial[5]

[1] Cogger, 1975.
[2] Wager, 1965.
[3] Personal observation.
[4] McDiarmid, 1971.
[5] Inger, pers. obs.
[6] Stebbins, 1966.

and each frog of each species was jumped during one time period out of each day for a four-day interval. Division of individuals into four groups, time of initial jumping sequence, and order of individuals jumped within a time period were obtained from a random numbers table.

The jumping ability of an individual frog was taken as the maximum distance jumped over the four-day test period from the total of 40 jumps made. The maximum distance was chosen rather than mean distance jumped because other studies have shown differential exhaustion rates among frogs (Rand, 1952; Bennett and Licht, 1974). If the mean of a large number of consecutive jumps is used (as in Wermel, 1934) frog species with rapid exhaustion rates may have an artificially lowered jumping ability compared to species with lower rates of fatigue. Animals were weighed after each jump sequence. After jump experiments were completed, all animals were anesthetized and body length and hindlimb length were measured.

Standard trajectory formulae (see Alexander, 1968) were used to calculate acceleration of a jump. The angle of take-off was assumed to be 45° for the maximum jump of each individual.

RESULTS

Body Geometry

Hindlimb:Snout-vent length

a. *Interspecific.*—Figure 2 shows the mean hindlimb:snout-vent ratio for 27 species of adult frogs. The graph includes data from this study and from Dobrowolska (1973). There is an association between predominant form of locomotion of each species and its hindlimb:snout-vent ratio (see Table 1). Those species with the lowest ratios tend toward walking exclusively, those with intermediate ratios hop, walk and burrow, and those with the highest ratios are exclusively jumpers. Notice that there is no correlation between mean adult size and hindlimb:snout-vent ratio, i.e., there is no regular change

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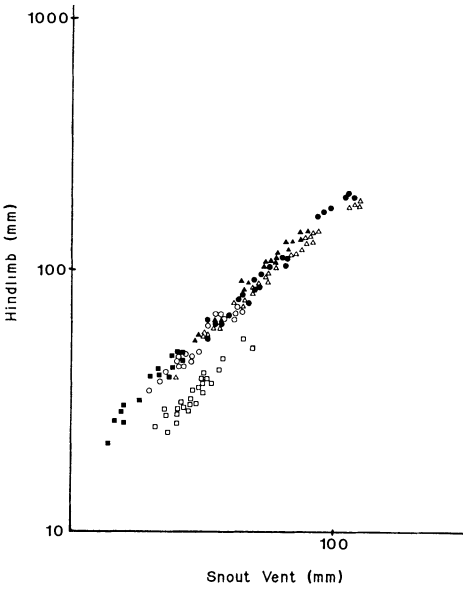


FIG. 3. Log-log plot of hindlimb length vs. snout-vent length in six species of frogs. White squares, *Breviceps adspersus*; black squares, *Hyperolius viridiflavius*; black triangles, *Rana erythracea*; black circles, *Rana catesbeiana*; white circles, *Hyla regilla*; white triangles, *Bufo asper*.

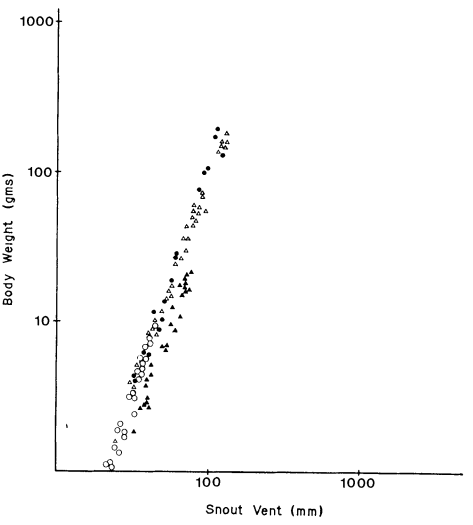


FIG. 4. Log-log plot of body weight vs. snout-vent length. Black triangles, *Rana erythracea*; black circles, *Rana catesbeiana*; white circles, *Hyla regilla*; white triangles, *Bufo asper*.

in hindlimb:snout-vent ratio with interspecific differences in body size.

Figure 3 is a log-log graph of the relationship between hindlimb and snout-vent length in ontogenetic series of six species (museum specimens).

Differences in relative hindlimb length of similar size (snout-vent length) species are reflected in the significant differences in the y-intercepts of their respective axes or the significant differences in slope (Appendix I*A*). These differences in y-intercept and slope confirm that there are differences in shape among frog species.

b. *Intraspecific.*—For five of the six species there is no significant difference between the calculated regression line slopes and isometry (predicted slope of 1.0) using the 95% confidence interval (Appendix I*A*). *Bufo asper* shows negative allometry in hindlimb length vs. snout-vent length, i.e., hindlimbs become relatively shorter with increasing body length.

Body weight:Snout-vent length

a. *Interspecific.*—Figure 4 shows the relationship between body weight and snout-vent length in four species of preserved animals: *Bufo asper*, *Hyla regilla*, *Rana catesbeiana*, and *Rana erythracea*. There are significant differences in weight among species of the same snout-vent length as indicated by differences in y-intercepts (see Appendix I*B*).

b. *Intraspecific.*—There are no significant differences between calculated regression slopes and isometry (predicted slope of 3.0) for three of the species. *Bufo asper*, however, does show negative allometry in body weight vs. snout-vent (Appendix I*B*).

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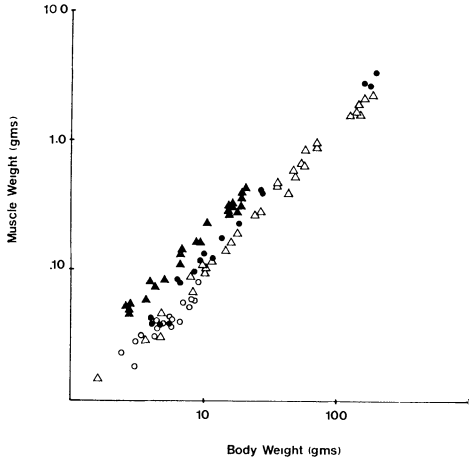


FIG. 5. Log-log plot of extensor muscle weight vs. body weight. Black triangles, *Rana erythracea*; black circles, *Rana catesbeiana*; white circles, *Hyla regilla*; white triangles, *Bufo asper*.

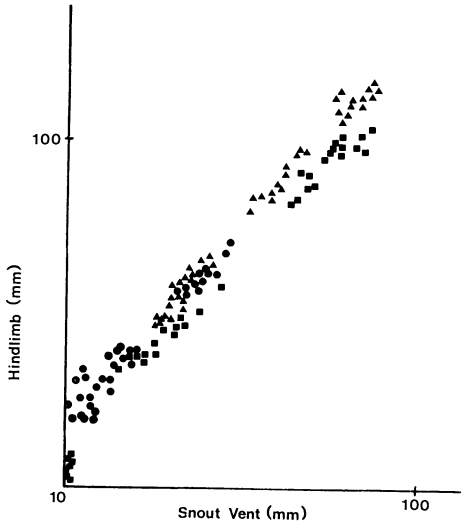


FIG. 6. Log-log plot of hindlimb length vs. snout-vent length for experimental animals over 10-mm snout-vent length. Black squares, *Bufo americanus*; black circles, *Pseudacris triseriata*; black triangles, *Rana pipiens*.

Extensor muscle weight:Body weight

a. *Interspecific.*—There are marked differences in hindlimb extensor muscle weight among *B. asper*, *R. catesbeiana*, *R. erythracea*, and *Hyla regilla*, as indicated by significant differences in *y*-intercepts and/or slope exponents (Fig. 5 and Appendix IC).

b. *Intraspecific.*—Extensor muscle weight scales isometrically in *H. regilla* and *R. erythracea*, but shows positive allometry ($a > 1.0$) in *R. catesbeiana* and *B. asper* (Fig. 5 and Appendix IC).

Locomotor Behavior

Body geometry of experimental animals

1. *Hindlimb:Snout-vent length.*—Figure 6 and Table 2 show the relationship between hindlimb and snout-vent length in *P. triseriata*, *B. americanus*, and *R. pipiens*, the species used in the jumping experiments. In this graph an additional size class has been added to the samples compared to museum specimens, that of "just metamorphosed" individuals (metamorphosis to two months). In the three species measured, the "just

TABLE 2. *The mean hindlimb/snout vent ratios with 95% confidence intervals for three size classes of* Pseudacris triseriata, Bufo americanus *and* Rana pipiens.

	$\overline{m}$HL/SV	95% Confidence interval
Pseudacris triseriata (Illinois)		
A. adults (n = 24)	1.64	±.036
B. juveniles (n = 10)	1.61	±.041
C. "Just metamorphosed" (n = 25)	1.54	±.039
Bufo americanus		
A. adults (n = 40)	1.58	±.037
B. juveniles (n = 17)	1.40	±.039
C. "Just metamorphosed" (n = 10)	1.11	±.048
Rana pipiens		
A. adults (n = 14)	1.88	±.080
B. juveniles (n = 16)	1.88	±.053
C. "Just metamorphosed" (n = 17)	1.67	±.049

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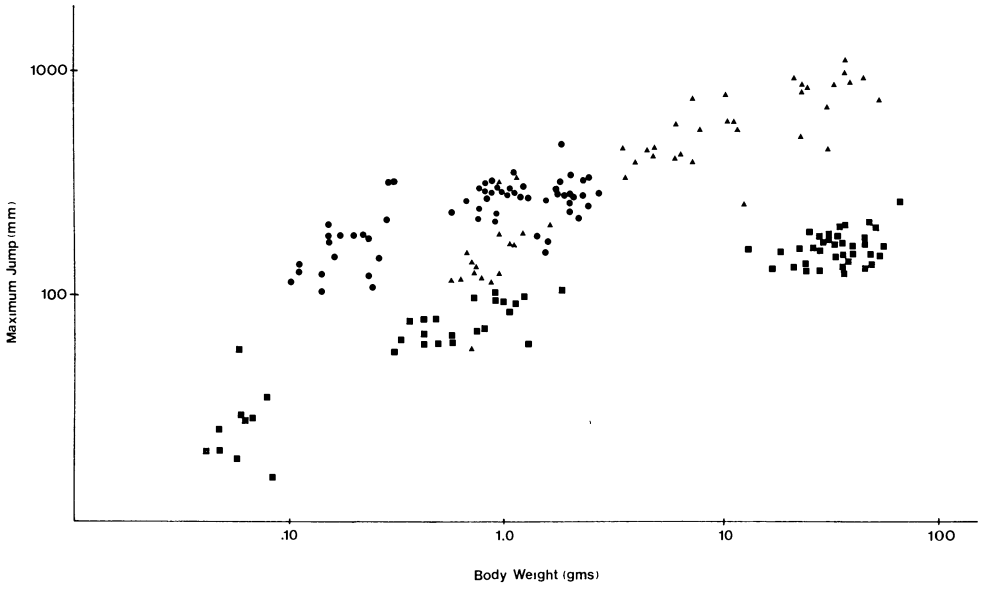


FIG. 7. Log-log plot of maximum jump vs. body weight in experimental animals. Black squares, *Bufo americanus*; black circles, *Pseudacris triseriata*; black triangles, *Rana pipiens*.

metamorphosed" frogs have lower hindlimb:snout-vent ratios than the adults (Table 2). The differences between young and adult hindlimb: snout-vent ratios indicate positive allometric growth of the hindlimbs of the three species for a period of time after metamorphosis, because within adults of a species the relationships are isometric. Similar patterns of positive allometric growth of the hindlimb have been found in some immature European anurans (Terentiev, 1943; Dobrowolska, 1973).

2. *Bodyweight : Snout-vent.* — When weights of live animals from the jumping experiments are regressed against their snout-vent lengths (Appendix II*B*), two species (*Rana pipiens* and *Bufo americanus*) show a positive allometry in weight ($a > 3.0$ at 95% confidence limit). This is in contrast to the museum specimens weighed, where none of the species showed positive allometry, and one species showed negative allometry. This deviation from museum specimen data may result from differential water retention among the live frogs.

Locomotor ability

Figure 7 shows the relationship between distance of maximum jump and body weight for the three experimental species, from metamorphosis through sexual maturity. The relationship between maximum jump and body weight is not linear when the "just metamorphosed" size classes are included in the sample. Note the marked differences in distance of jump between species of the same size (body weight).

Figure 8 and Appendix II*C* show the relationship between distance of maximum jump and body weight for juveniles and adults of the three species. Note that the slope of the regression line is greater than zero in all cases, but less than one. Frogs

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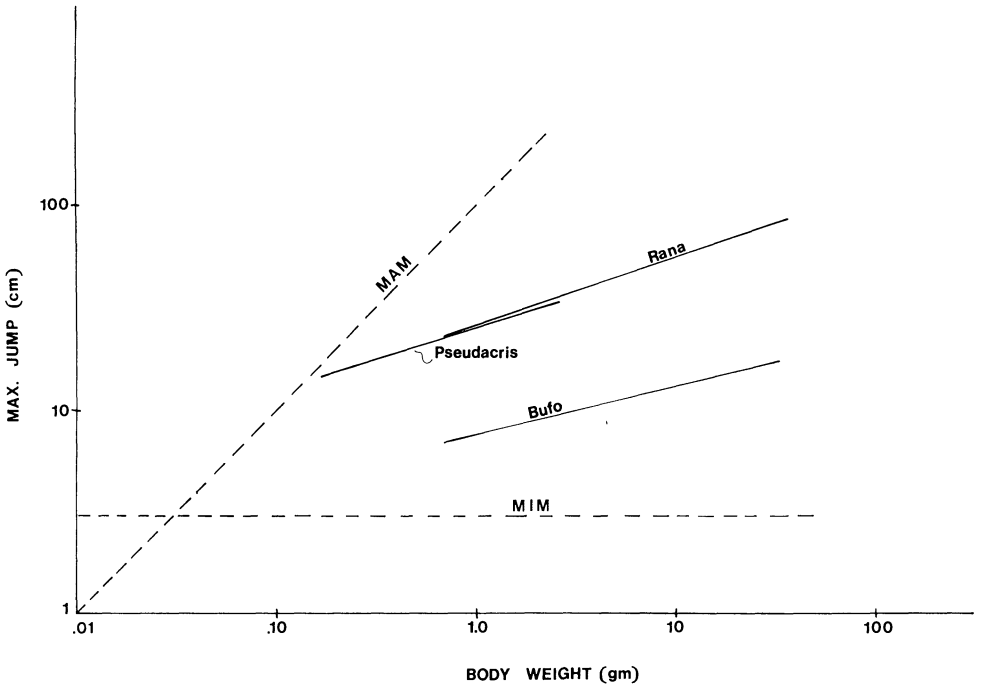


FIG. 8. Log-log plot of maximum jump vs. body weight compared to model predictions for morphological isometry (MIM) and morphological allometry (MAM).

jump longer absolute distances with increasing size, but shorter relative distances (absolute distance:body weight).

Figure 9 illustrates the relationship between acceleration and body weight for juveniles and adults of the three experimental species. Acceleration is roughly constant with increasing size in *Rana pipiens* and *Pseudacris triseriata*. Acceleration decreases with increasing size in *Bufo americanus*, but not as steeply as predicted by the morphological isometry model (Appendix II*D* and Fig. 9). The predicted slope of acceleration:body weight for the morphological isometry model is –0.33 (derived from standard trajectory formulae); the slope of acceleration: body weight for *Bufo americanus* is –0.15.

DISCUSSION

Analysis of acceleration and distance performance for jumping frogs reveals that: (1) all species tested increase jump distance with increasing size; (2) all species decrease relative jump distance with increasing size; (3) some species do maintain constant accelerations with increasing size; and (4) there are significant differences in acceleration and jump distance among frog species. These results indicate a need to (1) re-examine what the principal function of jumping is in frogs; (2) to explain the discrepancies between model predictions and parameter performances; and (3) make interspecific comparisons of locomotor morphology and physiology in an attempt to account for differences in performance among species.

Jump Function

Maintenance of constant acceleration with size increase requires morphological

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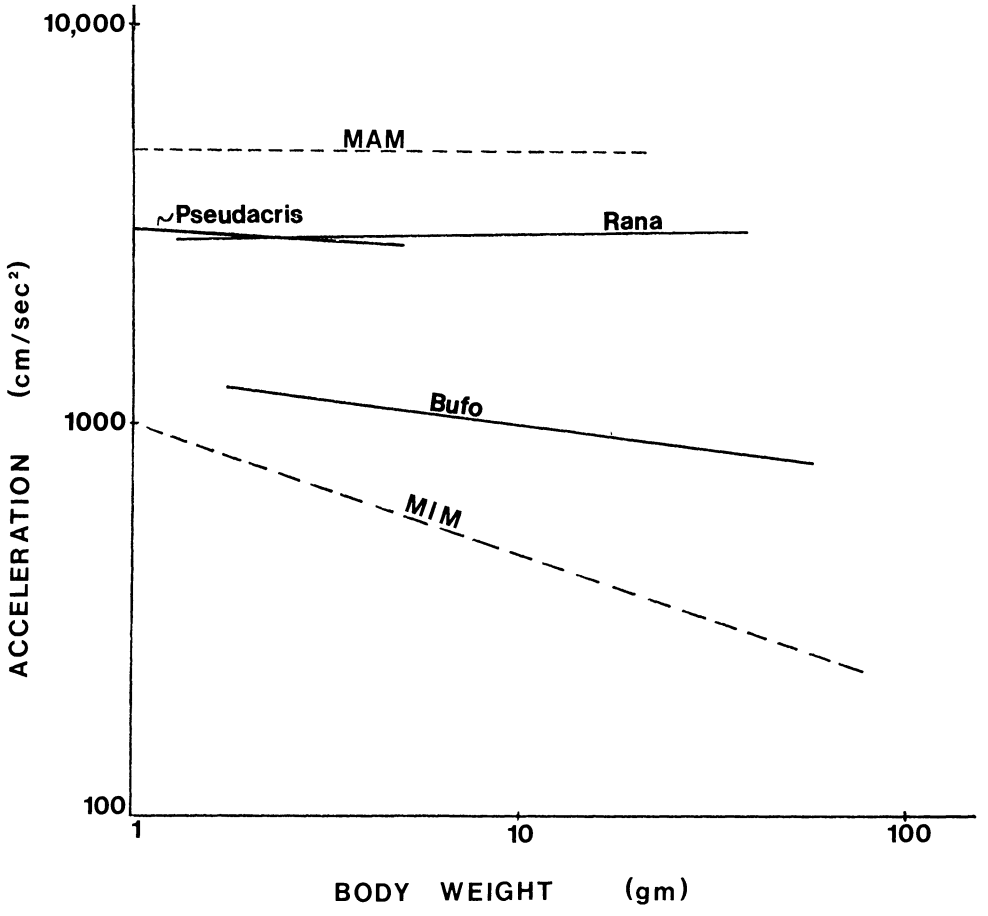


FIG. 9. Log-log plot of acceleration vs. body weight compared to model predictions for morphological isometry (MIM) and morphological allometry (MAM).

specialization and is energetically expensive in large animals. The fact that frogs of two species maintain high acceleration with increasing size suggests that quickness of movement may be a critical parameter of jumping for at least some frog species.

In general, smaller muscles and the muscles of smaller animals are quicker (Hill, 1950). Geometrically similar animals of different sizes are postulated to jump the same distance because in the larger animals slower muscle contraction time (relatively lower accelerations) is compensated for by longer take-off time. If larger animals maintain high accelerations, they must be maintaining rapid contraction of the hind-limb extensor musculature. Rapid contraction can be maintained against a varying load by adjusting the number of muscle fibers employed, but it exacts a metabolic cost. Preservation of a high intrinsic speed of contraction entails proportionally higher expenditure of energy in larger animals (Hill, 1950). If larger frogs are maintaining high accelerations by maintaining high speed of muscle contraction, one would expect to see differences in the amount of energy expenditure between large and small animals. No comparisons of energy expenditure between different sized individuals of the same species are available, but data from physio-

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logical work comparing species of different sizes suggest that this indeed may be the case. In lactate production studies on amphibians, Bennett and Licht (1974) found that exhaustion rates differed between two species of good jumpers, *Rana pipiens* (body weight 39.7 g) and *Hyla regilla* (body weight 2.6 g). After 10 min of activity, *Hyla regilla* was still capable of slow, uncoordinated jumps, while *Rana pipiens* was unable to sustain any degree of activity. In other words, jumping was energetically more expensive for the larger *Rana pipiens*.

Model and Predictions and Parameter Performance

Ontogenetically, most of the frog species measured in this study appear to maintain geometric similarity in body weight, body length, length of the hindlimbs, and weight of the hindlimb extensor muscles (Appendix I and II). This isometry is consistent with the morphological isometry model; consequently, one would expect frogs of a given species, regardless of size, to jump the same absolute distance, with larger frogs having smaller accelerations. Yet, jump experiments reveal that, within a species, larger frogs jump absolutely longer distances than do small frogs, and in two species, *Pseudacris triseriata* and *Rana pipiens*, nearly constant acceleration is maintained, despite increases in body size (Figs. 8 and 9). While body proportions within each frog species satisfy the assumption of the morphological isometry model, jump distance and acceleration conform more closely to the predictions of the morphological allometry model.

There are two possible explanations of these findings: (1) Hill's model is wrong or (2) an aspect of locomotor morphology or physiology not measured in this study is appropriately allometric.

Predictions of parameter behavior for quadrupedal locomotion, based on Hill's morphological isometry model, have been tested experimentally (Heglund, McMahon, and Taylor, 1974). The predictions have not held up for quadrupedal animals over a large size range. Subsequent work (McMahon, 1975), however, indicates that quadrupedal animals violate the basic assumption of the model: geometric similarity. We know that this is not the case for frogs, at least in the body proportions that were measured. Predictions of parameter behavior from Hill's model for jumping locomotion have been studied most thoroughly in arthropods (Bennet-Clark, 1975; Evans, 1973; Bennet-Clark and Lucey, 1967). Recently, the results of this work have been compared with data from vertebrates in a discussion of scale effects in jumping animals (Bennet-Clark, 1977). Interspecific comparisons of jumping animals show that not all animals jump the same height. For example, a locust cannot jump as high as a lesser galago (see Fig. 3, p. 193 Bennet-Clark, 1977). But, again, in this comparison, the basic assumption of Hill's model, geometric similarity, is violated. In locusts only 5% of the body weight is hindlimb extensor musculature, while in lesser galagos, hindlimb extensor musculature comprises 16% of the body weight (Bennet-Clark, 1977). In every case where predictions of Hill's model have not been borne out, experimentally, the basic assumption of the model has been violated. Could this be the case for frogs as well? Is there some aspect of locomotor morphology, unmeasured in this study, that is scaling allometrically?

The distance (or height) an animal jumps is ultimately determined by the energy per unit mass produced by the muscles (see Bennet-Clark, 1977 for this derivation). Thus, if all animals have the same proportion of muscle (i.e., are geometrically similar), and that muscle is capable of producing the same specific energy, they should all jump the same distance. Measurements of frogs indicate that muscle mass scales isometrically intraspecifically (except for very large species), but no direct measurements were made to determine scaling of energy. If

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TABLE 3. *The mean hindlimb/snout vent ratios and mean maximum jumps with 95% confidence intervals for adult* Pseudacris triseriata *(Illinois);* Pseudacris triseriata *(Colorado);* Bufo americanus *(Illinois).*

Species	N	Mean ratio HL/SV	95% Confidence interval	Mean max jump (MM)	95% Confidence interval
Pseudacris triseriata (Illinois)	24	1.64	±0.036	278.38	±18.45
Pseudacris triseriata (Colorado)	20	1.56	±0.035	281.70	±34.00
Bufo americanus	40	1.58	±0.037	165.30	±14.96

the specific energy produced by the hindlimb muscles of frogs changes with increasing size, deviations in acceleration and distance behavior from Hill's predictions could be accounted for. One indication that this may be the case is that frogs increase the number of muscle fibers per muscle unit mass with increasing size (D. Sperry, unpublished observations). Recruitment of additional muscle fibers with increasing size could change the energy per unit mass relationship and result in the intraspecific maintenance of constant acceleration and increasing jump distance seen in frogs in the jump experiments. Similarly, if hindlimb muscles of different sized frogs were composed of different proportions of muscle fiber types, the energy per unit mass relationship could change.

Interspecific Differences in Jumping Ability

Previous workers have found a positive correlation between the relative length of the hindlimbs and the length of jump among frog species. This correlation has led to the most generally accepted explanation for differences in jump distance among frog species, that it is related to differences in relative lengths of the hindlimbs (e.g., Zug, 1972). The mechanical basis for this explanation is that longer hindlimbs increase take-off velocity by increasing the distance through which the propulsive force acts (Gray, 1968). Results from the jump experiments of this study, however, show this difference in body proportions among species is not sufficient, in all cases, to account for significant differences in jumping ability (see Table 3). *Pseudacris triseriata* from Colorado have a significantly lower hindlimb: snout-vent ratio than *Pseudacris triseriata* from Illinois, but there is no significant difference in their jump distances. *Bufo americanus* jumps significantly shorter distances than individuals of both the Illinois and Colorado populations of *Pseudacris triseriata* but does not have a significantly lower hindlimb:snout-vent ratio.

Data from this study and work on other jumping animals (Bennet-Clark, 1977) suggest at least two other aspects of morphology that may be critical to understanding differential jumping ability among frog species. One factor is related to the relative amount of hindlimb extensor musculature. Frog species of the same body weight and same relative hindlimb length may differ significantly in amount of hindlimb extensor musculature (see Fig. 5). Thus, as a working hypothesis, interspecific differences in jump distance may be correlated as plausibly with differences in hindlimb musculature as with differences in the relative length of the hindlimbs.

A second factor is the use of stored energy in jumping. Bennet-Clark (1977) has shown that in animals with a body length under 1 m, more power may be used in a jump than can be supplied directly by the muscles. The extra power may come from energy stored in stretched tendons and ligaments. No studies have

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been done on energy storage for jumping in frogs, but from their small size and good jumping ability, it is likely that they too store energy for high power output. In addition to leg muscle tendons, frogs possess another possible energy storage reserve: the articular ligaments of the iliosacral joint. The frog pelvic girdle is bound to the vertebral column by ligaments which are under tension when the frog is in the pre-jump posture (personal observation to be documented). These ligaments are composed, in part, of elastin, a good material for energy storage (Bennet-Clark, 1977). The structure of the pelvic girdle ligaments and of the iliosacral articulation varies among frog species (Whiting, 1961; Emerson, to be documented). If these ligaments do function as energy stores, differences in their morphologies may relate to differences in their capacity to store energy, another possible contributing factor to the observed differences in jumping ability among frog species.

These two examples show that work to date on interspecific differences in jump distance among frog species is incomplete. A much finer understanding of more aspects of frog locomotor morphology is necessary before the causes of differential jumping ability can be established.

Summary

Analysis of gross morphology, by itself, is not sufficient to explain why frogs can maintain constant acceleration and increase jump distance with increasing size. Jumping ability is determined in part by properties of hindlimb extensor musculature that have not yet been adequately explored, such as physiological and architectural parameters.

Data from this study suggest that frog morphology facilitates maintenance of constant acceleration over a large size range, and that quickness of movement is a critical parameter of the jump as a predator escape mechanism.

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APPENDIX I

Museum specimen measurements

Species	Number of animals	Equations	95% Confidence intervals: Slope	95% Confidence intervals: y-intercept	Correlation coefficients
A. Hindlimb length/snout vent length					
Breviceps adspersus	26	$\log y = .900 \log x + .2077$	0.171	0.013	0.92
Hyperolius viridiflavius	18	$\log y = 1.110 \log x + .1016$	0.208	0.022	0.97
Rana erythracea	28	$\log y = .961 \log x + .3151$	0.097	0.010	0.97
Rana catesbeiana	18	$\log y = 1.023 \log x + .1985$	0.068	0.013	0.99
Hyla regilla	27	$\log y = 1.018 \log x + .2059$	0.040	0.004	0.99
Bufo asper	38	$\log y = .929 \log x + .3496$	0.032	0.006	0.99
Significant difference between slopes: *Rana catesbeiana* and *Bufo asper* ($t = 3.0575$); *Bufo asper* and *Hyla regilla* ($t = 2.7563$); *Breviceps adspersus* and *Hyperolius viridiflavis* ($t = 2.3856$).					
B. Body weight/snout vent length					
Rana erythracea	28	$\log y = 2.864 \log x - 4.1$	0.318	0.037	0.97
Rana catesbeiana	18	$\log y = 3.070 \log x - 4.1$	0.258	0.038	0.99
Hyla regilla	27	$\log y = 3.180 \log x - 4.3$	0.181	0.017	0.97
Bufo asper	38	$\log y = 2.828 \log x - 3.7$	0.134	0.021	0.98
Significant difference between slopes: *Hyla regilla* and *Bufo asper* ($t = 2.2666$).					
C. Extensor muscle weight/body weight					
Rana erythracea	24	$\log y = 1.03 \log x - 1.75$	0.051	0.016	0.99
Rana catesbeiana	15	$\log y = 1.12 \log x - 2.03$	0.064	0.034	0.99
Hyla regilla	16	$\log y = .841 \log x - 2.00$	0.201	0.033	0.92
Bufo asper	33	$\log y = 1.111 \log x - 2.10$	0.061	0.033	0.99
Significant difference between slopes: *Hyla regilla* and *Rana catesbeiana* ($t = 2.68$); *Hyla regilla* and *Rana erythracea* ($t = 2.22$); *Rana erythracea* and *Rana catesbeiana* ($t = 2.29$).					

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APPENDIX II

Experimental Animals (two classes: juveniles and adults)

Species	Number of animals	Equations	95% Confidence intervals		Correlation coefficients
			$+ayx$	$+byx$	
A. Hindlimb length/snout vent length					
Pseudacris triseriata	54	$\log y = 1.030 \log x + .1469$	0.070	0.0084	0.98
Bufo americanus	57	$\log y = 1.085 \log x + .0430$	0.038	0.0113	0.99
Rana pipiens	30	$\log y = 1.020 \log x + .5512$	0.102	0.0089	0.97
B. Body weight/snout vent length					
Pseudacris triseriata	54	$\log y = 3.080 \log x - 3.3979$	0.155	0.018	0.98
Bufo americanus	57	$\log y = 3.370 \log x - 4.5230$	0.084	0.020	0.99
Rana pipiens	30	$\log y = 3.287 \log x - 4.5229$	0.138	0.015	0.99
C. Jump/body weight					
Pseudacris triseriata	54	$\log y = .277 \log x + 2.40$	0.077	0.029	0.72
Bufo americanus	57	$\log y = .183 \log x + 1.92$	0.031	0.025	0.84
Rana pipiens	30	$\log y = .313 \log x + 2.43$	0.120	0.044	0.72
D. Acceleration/body weight					
Pseudacris triseriata	54	$\log y = -0.06 \log x + 3115.30$	—	—	
Bufo americanus	57	$\log y = -0.15 \log x + 1339.65$	—	—	
Rana pipiens	30	$\log y = +0.02 \log x + 2935.10$	—	—	

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